LINGHAI LIU

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EDUCATION

Yale University 2023 - Present

Ph.D. in Statistics & Data Science

Brown University 2019 - 2023

B.S. in Applied Mathematics - Computer Science (*Honors*) | B.A. in Mathematics
Magna Cum Laude | Sigma Xi | Senior Prize in Computer Science

SELECTED RESEARCH EXPERIENCES

Improved Dimensional Dependence of Composite Log-Concave Sampling Sept 2025 -
Advisor: Sinho Chewi *Yale University*

- Improve the dimensional dependence of high-accuracy sampling for composite log-concave densities by implementing a restricted Gaussian oracle with independent Metropolis-Hastings proposals.
- Analyzing the convergence of the independent Metropolis-adjusted Markov chain and Gibbs sampling scheme using conductance arguments by controlling higher-order Rényi divergence.

Topological Representations for Neural Networks May 2022 - May 2023
Advisor: Stuart Geman *Brown University*

- Designed generative models capable of learning a richer collection of abstract relationships using a multinomial idealization of biological neurons while ensuring model hierarchy and reusibility.
- Developed sufficient statistics that exploit patterns of neuronal activities with topological structures such as cycles that can potentially be used to represent relationships.

Jacobian-free Backpropagation in Inverse Problems May 2022 - Jan 2024
Advisor: Samy Wu Fung *REU Researcher, Emory University*

- Compared classical optimization, standard feed-forward networks (Denoising CNN), Deep Unrolling methods, and implicit deep learning with applications to inverse problems in imaging.
- Trained implicit networks to denoise images using Jacobian-free Backpropagation with fixed memory costs, which yielded comparable results with current models.
- "Training Implicit Networks for Image Deblurring using Jacobian-Free Backpropagation." arXiv preprint arXiv:2402.02065 (2024).

TEACHING & MENTORING

Teaching Fellow at Yale University

- S&DS4320/S&DS6320: Advanced Optimization Techniques (TF) Spring 2026
- S&DS4000/S&DS6000/MATH3300: Advanced Probability (TF) Fall 2025
- S&DS242/S&DS542/MATH242: Theory of Statistics (TF) Spring 2025
- S&DS230/S&DS530/ENV757/PLSC530: Data Exploration and Analysis (TF) Fall 2024

Teaching Assistant at Brown University (*graduate level)

- CSCI1952Q: Algorithmic Aspects of Machine Learning (TA) Spring 2023
- APMA1660: Statistical Inference II (TA) Spring 2023
- APMA1690: Computational Probability & Statistics (TA) Fall 2022

- APMA1740/2610*: Recent Applications in Probability & Statistics (TA) — Spring 2022
- DATA1010*: Probability, Statistics and Machine Learning (TA) — Fall 2021
- CSCI1470/2470*: Deep Learning (Head TA) — Fall 2021
- APMA1650: Statistical Inference I (TA) — Spring 2021

Peer Advisor, Meiklejohn Peer Advising Program — 2020 - 2021

- Advised 5 freshmen on academic and career development jointly with a faculty member.

HONORS & AWARDS

Emory REU/RET Computational Mathematics for Data Science, Emory University	2022
Second place, Henry Parker Manning Mathematical Prizes, Brown University	2022
BrownConnect SPRINT Undergraduate Teaching & Research Awards, Brown University	2021
Second place (tied), Hartshorn-Hypatia Freshman Math Contest, Brown University	2019

TECHNICAL STRENGTHS

Computer Languages	Python, Matlab, R, LaTeX, Stata, Java, C
Languages	English, Chinese, French

Date of Preparation: February 3, 2026